

Joint Strong Coordination and Sensing

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Abstract—We study a joint strong coordination and sensing scenario. A transmitter observes an independent and identically distributed (i.i.d.) $\bar{P}_U$ -source sequence U^N and communicates over a state-dependent memoryless channel $P_{YZ|XS}$ to a receiver that observes the Y -sequence and whose goal is to synthesize an action sequence V^N so that the joint law with U^N is close, in total variation, to an i.i.d. target law Q_{UV} . The channel provides generalized feedback Z to the transmitter (modelling signals backscattered from obstacles) from which the transmitter senses the hidden state S^N by means of the estimate $\hat{S}^N$ whose joint law with S^N should be close to an i.i.d. target law $Q_{S\hat{S}}$. Both terminals additionally share external common randomness of rate R_C . We first show that when R_C is sufficiently large, then the target laws $(Q_{UV}, Q_{S\hat{S}})$ can be strongly coordinated if, and only if, there exists an input pmf Q_X satisfying $I(U; V) \leq I(X; Y)$ and so that there exists a memoryless estimation kernel $Q_{\hat{S}|XZ}$ inducing the desired target law $Q_{S\hat{S}}$. A key step in the converse proof of this result is to establish optimality of such a memoryless estimation kernel. Our second result provides a sufficient condition on the available common randomness rate R_C so that the pair $(Q_{UV}, Q_{S\hat{S}})$ can be coordinated. This result is based on a feedback-aided joint source–channel block-Markov coding scheme including implicit common randomness extraction at the transmitter and the receiver.

I. INTRODUCTION

Recent years saw an emergence of integrated sensing and communication (ISAC) systems, in which the same resources are utilized for both communication and sensing tasks. Specifically, in these systems, a transmitter communicates to a receiver and, from the backscattered signals modelled as generalized feedback, attempts to estimate parameters of the environment. Information-theoretic studies on this subject include [1]–[6].

In this work, we take a *strong coordination* approach [7]–[11] to the ISAC problem. We consider the setup in Fig. 1, where the transmitter observes an independent and identically distributed (i.i.d.) U -sequence and the receiver produces a V -sequence such that the joint UV -sequence is strongly coordinated according to an i.i.d. target law Q_{UV} . Transmission takes place over a discrete-memoryless channel $P_{YZ|XS}$ governed by the i.i.d. state S . From the generalized feedback signal Z , the transmitter produces a state-estimate $\hat{S}$ so that the joint $S\hat{S}$ -sequence is strongly coordinated according to an i.i.d. target law $Q_{S\hat{S}}$.

We have two main contributions in this work. First we characterize (Theorem 1) the set of *all* target laws $(Q_{UV}, Q_{S\hat{S}})$

that can be coordinated over the channel $P_{YZ|XS}$ assuming that a sufficiently large common randomness rate R_C is available. In this case, coordination can be achieved with a simple separate source-channel coding scheme where the transmitter uses the generalized feedback not for coding but only for state estimation. Moreover, for state estimation, without loss in optimality, it can apply a simple memoryless estimation kernel $Q_{\hat{S}|XZ}$ to the input-feedback pairs. For situations where common randomness rate R_C is limited, our second result (Theorem 2) characterizes a subset of the target pairs $(Q_{UV}, Q_{S\hat{S}})$ that can be coordinated. This result is obtained with the output statistics of random binning (OSRB) approach [12] and based on a block-Markov coding scheme where the generalized feedback is also used for coding.

Similar coordination results over noisy channels have previously been obtained in [8], [13], [14], however all works do not include a sensing constraint. Moreover, [13] has no feedback, [8] considers weak rather than strong coordination, and [14] considers distortion and perception constraints. Similarly to our results, separate source-channel coding is optimal in [14] when sufficient common randomness is available.

Notation: We use standard notation, including $X^n := (X_1, \dots, X_n)$, we write $X^{i-1} := (X_1, \dots, X_{i-1})$, $X_{i+1}^n := (X_{i+1}, \dots, X_n)$, and $X_{\sim i} := (X^{i-1}, X_{i+1}^n)$. In the block-Markov proof, X_b^n denotes the length- n sequence in block b . For a given probability mass function (pmf) P , we denote by $P^{\otimes N}$ its N -fold product, N a positive integer, and for two pmfs P and Q we let $\|P - Q\|_{TV} = \sum_x |P(x) - Q(x)|/2$ denote their total variation (TV) distance. Finally, for any positive integer N , we write $[1 : N]$ for $\{1, \dots, N\}$.

II. SYSTEM MODEL AND PROBLEM FORMULATION

Consider the setup of Fig. 1. Node 1 observes a source block U^N , which is i.i.d. according to a given $\bar{P}_U$, and communicates over a state-dependent memoryless channel (SDMC) to Node 2, which then produces a coordinated sequence V^N at its end. Node 1 observes a feedback signal from the SDMC, based on which it produces an estimate $\hat{S}^N$ of an internal state sequence S^N . The SDMC is described by the transition law $\bar{P}_{YZ|XS}(y, z|x, s)$, meaning that at each time i and given channel input $X_i = x$ and internal state $S_i = s$, the channel output Y_i observed by Node 2 and the feedback output Z_i observed by Node 1 are jointly distributed according

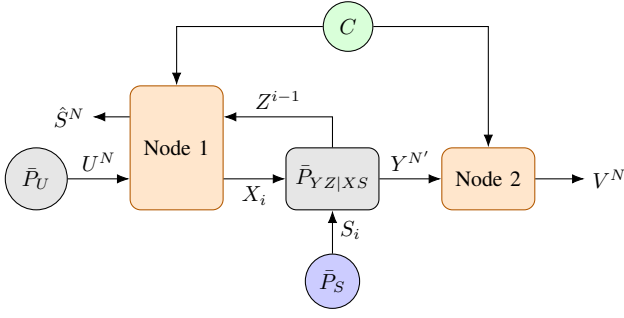


Fig. 1: Joint strong coordination and sensing setup.

to $\bar{P}_{YZ|XS}(\cdot, \cdot | x, s)$. The goal is that Node 2 coordinates its N outputs V^N with the N source symbols U^N according to a product pmf $Q_{UV}^{\otimes N}$ and that Node 1 coordinates its N estimates $\hat{S}^N$ with the first N internal state symbols S^N according to the product pmf $Q_{SS}^{\otimes N}$. To fully exploit the feedback, we allow the two nodes to employ the SDMC slightly more than N times. More precisely, we allow $N' \leq (1+\alpha)N$ channel uses, where α can be arbitrarily small but positive.

We assume that Node 1 observes U^N noncausally before transmission and has strictly causal access to the generalized feedback. We define the following notion of a code.

Definition 1. An (N, α, R_C) strong coordination–sensing code consists of:

- 1) Common-randomness C , independent of $(U^N, S^{N'})$ and uniform over $[1 : 2^{NR_C}]$, available to both nodes;
- 2) A sequence of stochastic encoding kernels

$\phi_i: \mathcal{U}^N \times \mathcal{Z}^{i-1} \times [1 : 2^{NR_C}] \rightarrow \mathcal{P}(\mathcal{X})$, $i \in [1 : N']$,
mapping source U^N , previous feedback outputs Z^{i-1} , and common randomness C to the i -th channel input X_i ;

- 3) A stochastic decoding kernel

$$g: \mathcal{Y}^{N'} \times [1 : 2^{NR_C}] \rightarrow \mathcal{P}(\mathcal{V}^N),$$

mapping channel outputs $Y^{N'}$ and C to a sequence V^N ;

- 4) A stochastic estimator kernel

$$h: \mathcal{U}^N \times [1 : 2^{NR_C}] \times \mathcal{X}^{N'} \times \mathcal{Z}^{N'} \rightarrow \mathcal{P}(\hat{\mathcal{S}}^N),$$

mapping U^N , C , and all channel inputs $X^{N'}$ and feedback outputs $Z^{N'}$ to the estimated sequence $\hat{S}^N$.

Our goal is to characterize the set of pmfs Q_{UV} and Q_{SS} that can be coordinated with a given rate of common randomness R_C and for given channel pmfs $\bar{P}_{YZ|XS}$ and $\bar{P}_S$.

Definition 2. A pair (Q_{UV}, Q_{SS}) is R_C -achievable if for any $\alpha > 0$ there exists a sequence (in N) of (N, α, R_C) strong coordination–sensing codes such that the induced joint pmfs on (U^N, V^N) and $(S^N, \hat{S}^N)$, denoted $P_{U^N V^N}$ and $P_{S^N \hat{S}^N}$ respectively, satisfy

$$\lim_{N \rightarrow \infty} \|P_{U^N V^N} - Q_{UV}^{\otimes N}\|_{TV} = 0, \quad (1)$$

$$\lim_{N \rightarrow \infty} \|P_{S^N \hat{S}^N} - Q_{SS}^{\otimes N}\|_{TV} = 0. \quad (2)$$

A pair (Q_{UV}, Q_{SS}) is simply called *achievable* if it is R_C -achievable for some finite $R_C \geq 0$.

III. MAIN RESULTS

Theorem 1 (Achievable Target Pairs). *A pair of target pmfs (Q_{UV}, Q_{SS}) is achievable if, and only if, there exist a pmf Q_X and a kernel $Q_{\hat{S}|XZ}$ such that the joint pmf*

$$Q_{UVSXYZ\hat{S}} = Q_{UV} Q_X \bar{P}_S \bar{P}_{YZ|XS} Q_{\hat{S}|XZ} \quad (3)$$

has $(S, \hat{S})$ -marginal Q_{SS} and satisfies

$$I(U; V) \leq I(X; Y). \quad (4)$$

The achievability part of above theorem can be proved with a joint source–channel coding scheme that uses the generalized feedback only for state estimation but not for coding. The estimation sequence is produced by applying a memoryless kernel to the inputs and generalized feedback symbols.

Theorem 2 (Sufficient Condition for R_C -Achievability). *A pair of pmfs (Q_{UV}, Q_{SS}) is R_C -achievable if there exists a pmf Q_{W_1} and kernels $Q_{X|UW_1}$, $Q_{W_2|UZ\hat{S}W_1}$, $Q_{V|YW_1W_2}$, and $Q_{\hat{S}|XZ}$ such that the joint pmf*

$$Q_{UVSXYZ\hat{S}W_1W_2} = \bar{P}_U \bar{P}_S Q_{W_1} Q_{X|UW_1} \bar{P}_{YZ|XS} \cdot Q_{W_2|UZ\hat{S}W_1} Q_{V|YW_1W_2} Q_{\hat{S}|XZ} \quad (5)$$

satisfies the following three conditions:

- 1) Its (U, V) - and $(S, \hat{S})$ -marginals are Q_{UV} and Q_{SS} .
- 2) The following information constraint is satisfied:

$$I(U, X, Z; W_2 | W_1, Y) \leq I(W_1; Y). \quad (6)$$

- 3) The rate of common randomness satisfies

$$R_C > H(W_1 W_2 | Y) - \min \left\{ \frac{H(W_2 | U, V)}{H(W_2 | S, \hat{S})} \right\}. \quad (7)$$

This sufficient condition is obtained with a block-Markov coding scheme, as described in Section V. Intuitively, in each block the W_1 -sequence is used to describe the current source sequence U to Node 2, while the W_2 -sequence is generated from the source, input, and the feedback variables of the previous block. This W_2 -sequence allows, on the one hand, refining the description of the previous block and, on the other, extracting common randomness from the channel, so as to allow for a smaller R_C . The two terms in the minimum in (7) quantify this extraction: $H(W_2 | U, V)$ is the binning randomness that can be recycled without disturbing the communication target and $H(W_2 | S, \hat{S})$ is the corresponding quantity for the sensing target. The estimated sequence $\hat{S}^N$ is produced by applying the memoryless kernel $Q_{\hat{S}|XZ}$ symbolwise to the input–feedback pairs (X^N, Z^N) .

Example 1. Let $U \sim \text{Bern}(1/2)$ and $S \sim \text{Bern}(1/2)$ be independent, and let the channel to the receiver and the generalized feedback be

$$Y = X \oplus N_Y(S), \quad N_Y(s) \sim \text{Bern}(p_s), \quad (8)$$

$$Z = S \oplus N_Z(X), \quad N_Z(x) \sim \text{Bern}(q_x), \quad (9)$$

with $(p_0, p_1) = (0.01, 0.19)$ and $(q_0, q_1) = (0.01, 0.49)$. We focus on binary symmetric target laws Q_{UV} and $Q_{S\hat{S}}$, i.e.,

$$Q_{V|U}(v|u) = (1 - D_c)\mathbf{1}\{v = u\} + D_c\mathbf{1}\{v \neq u\}, \quad (10)$$

$$Q_{\hat{S}|S}(\hat{s}|s) = (1 - D_s)\mathbf{1}\{\hat{s} = s\} + D_s\mathbf{1}\{\hat{s} \neq s\}, \quad (11)$$

for $D_c, D_s \in [0, 1]$. Figure 2 shows the (D_c, D_s) achievable region by Theorem 2 for common-randomness budgets $R_C \in \{0, 0.25, 1\}$ bits per source symbol, where we optimize over all admissible pmfs and kernels in Theorem 2. The curve $R_C = 1$ coincides with the condition in Theorem 1.

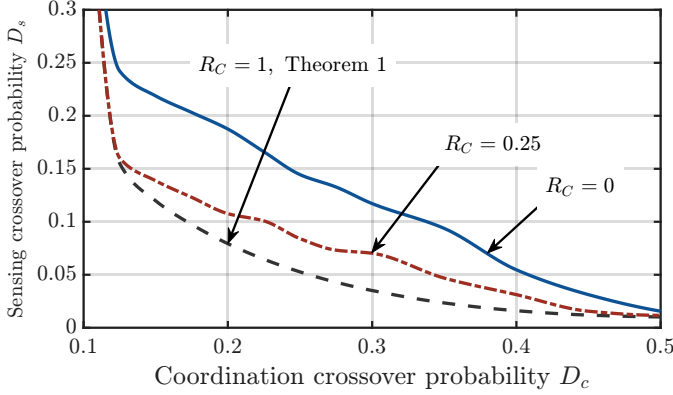


Fig. 2: Communication-sensing distortion region achieved by Theorem 2 for three common-randomness budgets.

IV. PROOF OF THEOREM 1

Achievability follows by specializing Theorem 2 to $W_1 = X$ and $W_2 = (X, V)$ with V independent of (X, Y, Z, S) .

Suppose that $(Q_{UV}, Q_{S\hat{S}})$ is R_C -achievable for some finite R_C , and fix $\alpha > 0$. Consider an (N, α, R_C) code with $N' \leq (1 + \alpha)N$. Let Ξ and Θ denote the internal local randomness employed at the stochastic encoder and decoder, so that X_i is a deterministic function of $(U^N, X^{i-1}, Z^{i-1}, C, \Xi)$ and V^N is a deterministic function of $(Y^{N'}, C, \Theta)$, with Θ independent of $(U^N, S^{N'}, C, \Xi)$. Let T be uniform over $[1 : N]$ and independent of everything, and define

$$\begin{aligned} U &\triangleq U_T, & V &\triangleq V_T, & X &\triangleq X_T, \\ S &\triangleq S_T, & Y &\triangleq Y_T, & Z &\triangleq Z_T, & \hat{S} &\triangleq \hat{S}_T. \end{aligned} \quad (12)$$

We bound $I(U^N; Y^{N'}|C)$ from below and above. Start with

$$\begin{aligned} I(U^N; Y^{N'}|C) &= I(U^N; C, \Theta, Y^{N'}) \geq I(U^N; V^N) \\ &\geq \sum_{i=1}^N I(U_i; V_i) \geq NI(U; V), \end{aligned} \quad (13)$$

where last inequality holds by the independence of U and T . On the other hand,

$$\begin{aligned} I(U^N; Y^{N'}|C) &= \sum_{i=1}^{N'} I(U^N; Y_i|Y^{i-1}, C) \\ &\leq \sum_{i=1}^{N'} I(U^N, X^{i-1}, Y^{i-1}, Z^{i-1}, C, \Xi; Y_i) \end{aligned}$$

$$\stackrel{(a)}{\leq} \sum_{i=1}^{N'} I(X_i; Y_i) \stackrel{(b)}{\leq} N(I_{Q_X^{(N)}}(X; Y) + \alpha \log |\mathcal{Y}|), \quad (14)$$

where (a) follows from the Markov chain $(U^N, X^{i-1}, Y^{i-1}, Z^{i-1}, C, \Xi) - X_i - Y_i$ and (b) follows from concavity of mutual information in the input pmf, where

$$Q_X^{(N)}(x) \triangleq \frac{1}{N} \sum_{i=1}^N P_{X_i}(x), \quad (15)$$

and from $N' - N \leq \alpha N$. Combining (13) and (14) gives

$$I_{P_{U_T V_T}}(U; V) \leq I_{Q_X^{(N)}}(X; Y) + \alpha \log |\mathcal{Y}|. \quad (16)$$

For this fixed α , compactness of the finite-dimensional probability simplices allows us to choose a subsequence $N_i \rightarrow \infty$ along which $Q_X^{(N_i)}$ converges to a pmf $Q_X^{(\alpha)}$. Strong coordination implies $P_{U_T V_T}^{(N_i)} \rightarrow Q_{UV}$, and continuity of mutual information therefore yields

$$I_{Q_{UV}}(U; V) \leq I_{Q_X^{(\alpha)}}(X; Y) + \alpha \log |\mathcal{Y}|. \quad (17)$$

The estimator can be treated along the same subsequence. App. B shows that one can select a kernel $Q_{\hat{S}|XZ}^{(\alpha)}$ so that

$$\begin{aligned} Q_{S\hat{S}}(s, \hat{s}) &= \sum_x Q_X^{(\alpha)}(x) \bar{P}_S(s) \\ &\quad \cdot \sum_{y,z} \bar{P}_{YZ|XS}(y, z|x, s) Q_{\hat{S}|XZ}^{(\alpha)}(\hat{s}|x, z). \end{aligned} \quad (18)$$

Finally, let $\alpha \downarrow 0$. By compactness, a subsequence of $(Q_X^{(\alpha)}, Q_{\hat{S}|XZ}^{(\alpha)})$ converges to $(Q_X, Q_{\hat{S}|XZ})$. Passing to the limit in (17) and (18) gives (4) and the required $(S, \hat{S})$ -marginal under (3). This completes the converse.

V. PROOF OF THEOREM 2

The proof of our sufficient condition is based on a block-Markov coding scheme. To this end, let $\alpha > 0$ and choose B sufficiently large such that $\frac{1}{B} < \alpha$. Then, split the blocklength N' into $B + 1$ blocks, each of length $n \triangleq \lfloor N'/B \rfloor$. Since $(B + 1)n \leq (\frac{B+1}{B})N' < N'$, some channel uses remain; these are left unused when (6) is strict and employed to handle equality. The transmitter describes each source block U_b^n during channel blocks b and $b + 1$, for $b = 1, \dots, B$, and the receiver produces the reconstruction sequence V_b^n based on output blocks Y_b^n and Y_{b+1}^n . The detailed encodings and decodings will be described shortly, see Subsections V-A and V-B. At the very end, the transmitter produces the estimation sequence $\hat{S}^N$ by employing a memoryless stochastic kernel $Q_{\hat{S}|XZ}$ to the input and feedback sequences (X^N, Z^N) . Analysis is based on the fact that the block-Markov procedure imposes a joint law of the following form, for some “hidden” random variables $\{H_{1,b}^n, H_{2,b}^n\}_{b=1}^B$ explained later:

$$\begin{aligned} P_{U^N, V^N, S^N, \hat{S}^N, H_{1,1}^n, \dots, H_{1,B}^n, H_{2,1}^n, \dots, H_{2,B}^n} \\ = P_{U_1^n, H_{1,1}^n} \prod_{b=2}^B P_{V_{b-1}^n, S_{b-1}^n, \hat{S}_{b-1}^n, H_{2,b-1}^n, U_b^n, H_{1,b}^n | U_{b-1}^n, H_{1,b-1}^n} \end{aligned}$$

$$\cdot P_{V_B^n, S_B^n, \hat{S}_B^n, H_{2,B}^n | U_B^n, H_{1,B}^n}. \quad (19)$$

Using a sliding-window analysis, we will show that

$$P_{U_1^n, H_{1,1}^n} = Q_{U, H_1}^{\otimes n}, \quad (20a)$$

and for $b = 2, \dots, B$:

$$\mathbb{E}_{Q_{U', H_1'}^{\otimes n}} \left[\left\| P_{V_{b-1}^n, S_{b-1}^n, \hat{S}_{b-1}^n, H_{2,b-1}^n, U_b^n, H_{1,b}^n | U_{b-1}^n, H_{1,b-1}^n}(\cdot | U'^n, H_1'^n) - \Gamma_{V, S, \hat{S}, H_2, U, H_1 | U', H_1'}^{\otimes n}(\cdot | U'^n, H_1'^n) \right\|_{\text{TV}} \right] \leq \eta_n. \quad (20b)$$

Here $\eta_n \rightarrow 0$ as $n \rightarrow \infty$ and

$$\Gamma_{V, S, \hat{S}, H_2, U, H_1 | U', H_1'} \triangleq Q_{V, S, \hat{S}, H_2 | U', H_1'} Q_{U, H_1},$$

so that the resulting per-block pmf

$$Q_{U, V, S, \hat{S}, H_1, H_2} = Q_{U, H_1} Q_{V, S, \hat{S}, H_2 | U, H_1}$$

has marginals $Q_{U, V}$ and $Q_{S, \hat{S}}$. The same bound, after marginalizing the newly generated $(U_b^n, H_{1,b}^n)$, applies to the termination block that produces $(V_B^n, S_B^n, \hat{S}_B^n, H_{2,B}^n)$. By the triangle inequality for TV distance (for details, see Appendix A-A) and the fact that marginalizing out can only reduce TV distance [15, Lemma 16], we then obtain over all blocks, with $N = Bn$:

$$\left\| P_{U^N, V^N} - Q_{U, V}^{\otimes N} \right\|_{\text{TV}} \leq (B+1)\eta_n, \quad (21a)$$

$$\left\| P_{S^N, \hat{S}^N} - Q_{S, \hat{S}}^{\otimes N} \right\|_{\text{TV}} \leq (B+1)\eta_n. \quad (21b)$$

Therefore, for any fixed B the TV distance vanishes. Letting $n \rightarrow \infty$, we conclude the proof for all values of α . It remains to explain the detailed encodings and decodings in each block and to prove Inequalities (20). We employ the output statistics of random binning approach (OSRB) to obtain the per-block encodings and decodings. In Subsection V-A we describe a random-binning scheme, sketched in Fig. 3, which implicitly defines two conditional distributions used as per-block encoders and decoders in our block-Markov scheme depicted in Fig. 4; see Subsection V-B.

A. Per-Block Random Binning Scheme

Consider the random-binning scheme in Fig. 3, where we arbitrarily choose Q_{W_1} and the kernels $Q_{X|UW_1}$, $Q_{W_2|UZXW_1}$, and $Q_{V|YW_1W_2}$, and define the averaged channel

$$\bar{P}_{Y|Z|X}(y, z|x) \triangleq \sum_{s \in \mathcal{S}} \bar{P}_S(s) \bar{P}_{Y|Z|XS}(y, z|x, s). \quad (22)$$

The coding scheme only depends on $P_{Y|Z|X}$ instead of $P_{Y|Z|XS}$ since no terminal directly observes the S -sequence.

We then define the target distribution

$$Q_{UW_1XYZW_2V} = \bar{P}_U Q_{W_1} Q_{X|UW_1} \bar{P}_{Y|Z|X} \cdot Q_{W_2|UZXW_1} Q_{V|YW_1W_2}. \quad (23)$$

For block 1, draw $W_{1,1}^n \sim Q_{W_1}^{\otimes n}$, $X_1^n \sim Q_{X|UW_1}^{\otimes n}(\cdot | U_1^n, W_{1,1}^n)$, and $(Y_1^n, Z_1^n) \sim \bar{P}_{YZ|X}^{\otimes n}(\cdot, \cdot | X_1^n)$. For blocks $b = 2, \dots, B$, the tuple from the previous block

$$\Sigma_{b-1}^n \triangleq (U_{b-1}^n, W_{1,b-1}^n, X_{b-1}^n, Y_{b-1}^n, Z_{b-1}^n) \quad (24)$$

and $(U_b^n, W_{1,b}^n)$ are fed to the chosen kernels to generate

$$\begin{cases} W_{2,b-1}^n & \sim Q_{W_2|UZXW_1}^{\otimes n}(\cdot | U_{b-1}^n, Z_{b-1}^n, X_{b-1}^n, W_{1,b-1}^n), \\ W_{1,b}^n & \sim Q_{W_1}^{\otimes n}, \\ X_b^n & \sim Q_{X|UW_1}^{\otimes n}(\cdot | U_b^n, W_{1,b}^n), \\ (Y_b^n, Z_b^n) & \sim \bar{P}_{YZ|X}^{\otimes n}(\cdot, \cdot | X_b^n). \end{cases}$$

For the termination block $b = B+1$, no new source block is coordinated. Thus draw $W_{1,B+1}^n \sim Q_{W_1}^{\otimes n}$ and $X_{B+1}^n \sim Q_{X|W_1}^{\otimes n}(\cdot | W_{1,B+1}^n)$ for

$$Q_{X|W_1}(x|w_1) = \sum_u \bar{P}_U(u) Q_{X|UW_1}(x|u, w_1), \quad (25)$$

and let $(Y_{B+1}^n, Z_{B+1}^n) \sim \bar{P}_{YZ|X}^{\otimes n}(\cdot, \cdot | X_{B+1}^n)$. In addition, for every $b = 1, \dots, B+1$, generate two random bin indices [12] of rates $\tilde{R}_F$ and $\tilde{R}_J$:

$$F_b = \varphi_{1,b}(W_{1,b}^n, W_{2,b-1}^n), \quad J_b = \varphi_{2,b}(W_{1,b}^n, W_{2,b-1}^n), \quad (26)$$

using independent random binnings $\varphi_{1,b}$ and $\varphi_{2,b}$, where $W_{2,0}$ is empty so that in block 1 only $W_{1,1}^n$ is binned. Based on these bin indices and the side information $(Y_b^n, W_{1,b-1}^n, Y_{b-1}^n)$, for every $b = 1, \dots, B+1$, a Slepian-Wolf decoder [12], [16] is employed to produce the reconstructions $(\hat{W}_{1,b}^n, \hat{W}_{2,b-1}^n)$ and finally generate

$$V_{b-1}^n \sim Q_{V|YW_1W_2}^{\otimes n}(\cdot | Y_{b-1}^n, W_{1,b-1}^n, \hat{W}_{2,b-1}^n). \quad (27)$$

For given binnings $\varphi_{1,b}, \varphi_{2,b}$, the above construction induces the following decoder and encoder kernels:

$$P_{\hat{W}_{1,b}^n, \hat{W}_{2,b-1}^n | F_b, J_b, W_{1,b-1}^n, Y_{b-1}^n}^{\text{SW}}, \quad (28)$$

$$P_{W_{1,b}^n, W_{2,b-1}^n | U_{b-1}^n, W_{1,b-1}^n, X_{b-1}^n, Z_{b-1}^n, F_b, J_b}^{\text{RB}}. \quad (29)$$

We refer to the joint induced pmf as P^{RB} .

B. Random Coding Scheme

Split the common randomness C into equally sized parts $(J_1, J_2, \dots, J_{B+1})$, and assume for the moment extra common randomness $(F_1, F_2, \dots, F_{B+1})$ of rate $\tilde{R}_F$. The randomness F_b , will be eliminated later on. We start by describing the encoding. In block $b = 1$, the encoder produces $W_{1,1}^n$ from (F_1, J_1) through the kernel $P_{W_{1,1}^n | F_1, J_1}^{\text{RB}}$, generates $X_1^n \sim Q_{X|UW_1}^{\otimes n}(\cdot | U_1^n, W_{1,1}^n)$, and transmits X_1^n over the channel; Node 2 reconstructs $\hat{W}_{1,1}^n$ from (F_1, J_1, Y_1^n) and uses it as side information in block 2. For blocks $b = 2, \dots, B$, the encoding is described in Figure 4, where we choose

$$P^{\text{Enc}} \triangleq P_{W_{1,b}^n, W_{2,b-1}^n | U_{b-1}^n, W_{1,b-1}^n, X_{b-1}^n, Z_{b-1}^n, F_b, J_b}^{\text{RB}}. \quad (30)$$

That means, based on the common-randomness bits F_b and J_b , and the sequences $(U_{b-1}^n, W_{1,b-1}^n, X_{b-1}^n, Z_{b-1}^n)$ generated in

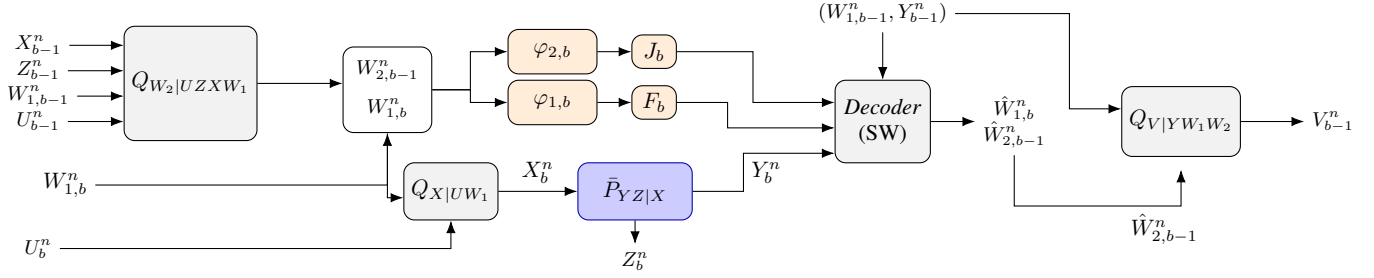


Fig. 3: Random-binning scheme for blocks $b \geq 2$.

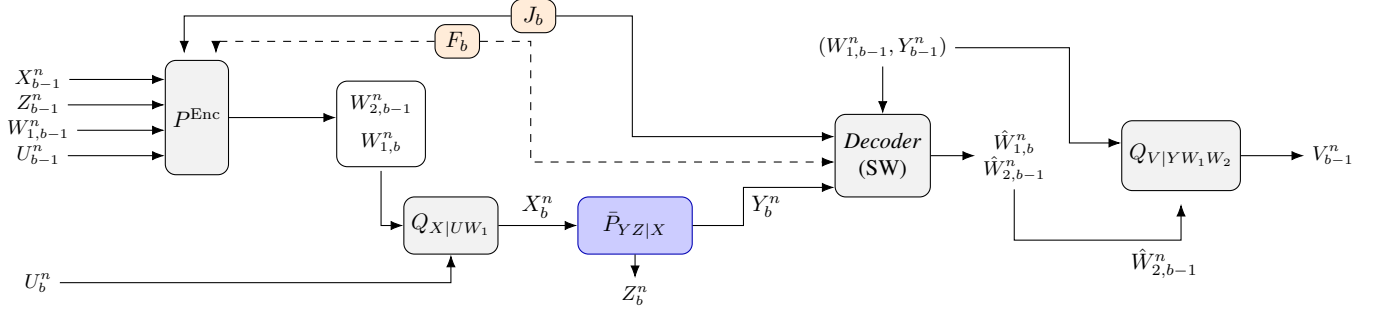


Fig. 4: Random-coding scheme for blocks $b \geq 2$. The dashed index F_b is initially treated as extra common randomness and is fixed later.

the previous block, it constructs the sequences $(W_{1,b}^n, W_{2,b-1}^n)$ and feeds $W_{1,b}^n$ and U_b^n to the kernel $Q_{X|UW_1}$ to construct $X_b^n \sim Q_{X|UW_1}^{\otimes n}(\cdot|U_b^n, W_{1,b}^n)$, which it then sends through the channel. In the termination block $b = B + 1$, the same encoder kernel produces $W_{1,B+1}^n$ and $W_{2,B}^n$, and the channel input is generated according to $Q_{X|W_1}^{\otimes n}(\cdot|W_{1,B+1}^n)$ as in (25). Figure 4 also describes the decoding steps at the end of each block $b = 2, \dots, B + 1$. The decoder first applies the kernel in (28) to (F_b, J_b) and $(W_{1,b-1}^n, Y_{b-1}^n, Z_{b-1}^n)$ to produce $(\hat{W}_{1,b}^n, \hat{W}_{2,b-1}^n)$. It finally generates

$$V_{b-1}^n \sim Q_{V|YW_1W_2}^{\otimes n}(\cdot|Y_{b-1}^n, W_{1,b-1}^n, \hat{W}_{2,b-1}^n).$$

We refer to the joint pmf induced above as P^{RC} .

C. TV Analysis

In a first instance, we assume that $(F_2, \dots, F_{B+1})$ is additional common randomness and show that when

$$\tilde{R}_F + \tilde{R}_J > H(W_1|Y) + H(W_2|W_1, Y), \quad (31)$$

$$\tilde{R}_F + \tilde{R}_J < H(W_1) + H(W_2|U, X, Z, W_1), \quad (32)$$

the pmf P^{RC} satisfies (20) for

$$H_{1,b}^n = (W_{1,b}^n, X_b^n, Y_b^n, Z_b^n), \quad H_{2,b}^n = W_{2,b}^n.$$

This is detailed in Appendix A-B, and is done in two steps:

- 1) Prove that (20) is satisfied by the relevant part of P^{RB} .
- 2) Prove that P^{RB} and P^{RC} are TV-vanishingly close.

Then, the triangle inequality concludes the proof. In block 1 only $W_{1,1}^n$ is binned, so that (31)–(32) specialize to $H(W_1|Y) < \tilde{R}_{F,1} + \tilde{R}_{J,1} < H(W_1)$, which is feasible by (6) and whose common-randomness cost $(\tilde{R}_{F,1} + \tilde{R}_{J,1})/B <$

$\alpha H(W_1)$ is arbitrarily small for B sufficiently large. Note that the analysis is performed on expectation over the random choice of the binning schemes. Vanishing TV distances on expectation ensure that the TV distances vanish for at least one realization of the binnings. Moreover, ensuring (20) coordinates the whole sequence of involved random variables, while the effective requirement to coordinate the U - and V -sequences and the S - and $\hat{S}$ -sequences is weaker. In Appendix A-C, we show that the extra randomness F^B can be fixed while preserving the desired strong coordination whenever

$$\tilde{R}_F < H(W_2|U, V), \quad \tilde{R}_F < H(W_2|S, \hat{S}). \quad (33)$$

After fixing F^B , only J^B consumes external common randomness. Combining these conditions with (31)–(32) gives (6) and (7); see Appendix A-D.

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APPENDIX A

DETAILS FOR ACHIEVABILITY PROOF

A. Proof Step in (21)

Let P_{AB} and Q_{AB} be distinct distributions. By the triangle inequality,

$$\begin{aligned} & \|P_{AB} - Q_{AB}\|_{TV} \\ & \leq \|P_{AB} - Q_A P_{B|A}\|_{TV} + \|Q_A P_{B|A} - Q_{AB}\|_{TV}. \end{aligned} \quad (34)$$

The first term is upper bounded by $\|P_A - Q_A\|_{TV}$, since total variation cannot increase under a fixed conditional kernel [15, Lemma 17]. For the second term,

$$\|Q_A P_{B|A} - Q_A Q_{B|A}\|_{TV} = \mathbb{E}_{Q_A} [\|P_{B|A} - Q_{B|A}\|_{TV}].$$

Combining the two bounds proves

$$\begin{aligned} \|P_{AB} - Q_{AB}\|_{TV} & \leq \|P_A - Q_A\|_{TV} \\ & + \mathbb{E}_{Q_A} [\|P_{B|A} - Q_{B|A}\|_{TV}]. \end{aligned} \quad (35)$$

B. Detailed Analysis of TV Distance

Let $\mathbf{P}^{\text{RC}}$ and $\mathbf{P}^{\text{RB}}$ denote the random pmfs, where the randomness stems from the binning functions $\varphi_{1,b}$ and $\varphi_{2,b}$, induced by our binning and coding schemes in Subsections V-A and V-B. Recall the definition of Σ_{b-1} in (24) and define

$$\Lambda_b^n \triangleq (W_{2,b-1}^n, V_{b-1}^n, U_b^n, W_{1,b}^n, X_b^n, Y_b^n, Z_b^n), \quad (36)$$

$$\tilde{\Lambda}_b^n \triangleq (\Lambda_b^n, \hat{W}_{1,b}^n, \hat{W}_{2,b-1}^n). \quad (37)$$

1) $\mathbf{P}^{\text{RB}}$ close to i.i.d.: To prove the first point, we define an intermediate distribution $\mathbf{P}^{\text{RB,perf}}$, which is induced by the auxiliary random-binning setup under perfect Slepian–Wolf decoding:

$$\begin{aligned} & \mathbf{P}_{F_b, J_b, \tilde{\Lambda}_b^n | \Sigma_{b-1}^n}^{\text{RB,perf}} \\ & \triangleq \mathbf{P}_{F_b, J_b | U_{b-1}^n, W_{1,b-1}^n, X_{b-1}^n, Z_{b-1}^n}^{\text{RB}} \\ & \cdot \mathbf{P}_{W_{1,b}^n, W_{2,b-1}^n | U_{b-1}^n, W_{1,b-1}^n, X_{b-1}^n, Z_{b-1}^n, F_b, J_b}^{\text{RB}} \\ & \cdot \bar{P}_{U_b^n} Q_{X_b^n | U_b^n, W_{1,b}^n} \bar{P}_{Y_b^n, Z_b^n | X_b^n} \\ & \cdot \mathbf{1}\{(\hat{W}_{1,b}^n, \hat{W}_{2,b-1}^n) = (W_{1,b}^n, W_{2,b-1}^n)\} \\ & \cdot Q_{V_{b-1}^n | Y_{b-1}^n, W_{1,b-1}^n, W_{2,b-1}^n}. \end{aligned} \quad (38)$$

Under this distribution, after marginalizing over the bin indices, we have

$$\mathbf{P}_{\Lambda_b^n | \Sigma_{b-1}^n}^{\text{RB,perf}} = Q_{\Lambda | \Sigma}^{\otimes n}, \quad (39)$$

where

$$\begin{aligned} Q_{\Lambda | \Sigma} & \triangleq Q_{W_2 | U Z X W_1} Q_{V | Y W_1 W_2} \\ & \cdot \bar{P}_{U'} Q_{W_1'} Q_{X' | U' W_1'} \bar{P}_{Y' Z' | X'}. \end{aligned} \quad (40)$$

Hence, if $\Sigma_{b-1}^n \sim Q_{\Sigma}^{\otimes n}$, where

$$\begin{aligned} Q_{\Sigma}(u, w_1, x, y, z) & \triangleq \bar{P}_U(u) Q_{W_1}(w_1) Q_{X | U W_1}(x | u, w_1) \\ & \cdot \bar{P}_{Y Z | X}(y, z | x), \end{aligned} \quad (41)$$

then

$$\mathbf{P}_{\Sigma_{b-1}^n \Lambda_b^n}^{\text{RB,perf}} = (Q_{\Sigma} Q_{\Lambda | \Sigma})^{\otimes n}. \quad (42)$$

It remains to prove that $\mathbf{P}^{\text{RB,perf}}$ is close to $\mathbf{P}^{\text{RB}}$ in total variation. Applying [17, Theorem 10.1] to the reconstruction of $(W_{1,b}^n, W_{2,b-1}^n)$, with side information $(W_{1,b-1}^n, Y_{b-1}^n, Y_b^n)$ and indices (F_b, J_b) available to the decoder, gives this conclusion if

$$\tilde{R}_F + \tilde{R}_J > H(W_1 | Y) + H(W_2 | W_1, Y). \quad (43)$$

More precisely, there exists a vanishing sequence $\delta_n \rightarrow 0$ such that

$$\mathbb{E}_{\varphi} \left[\mathbb{E}_{\mathbf{P}_{\Sigma_{b-1}^n}^{\text{RB}}} \left[\left\| \mathbf{P}_{\tilde{\Lambda}_b^n | \Sigma_{b-1}^n}^{\text{RB}} - \mathbf{P}_{\tilde{\Lambda}_b^n | \Sigma_{b-1}^n}^{\text{RB,perf}} \right\|_{TV} \right] \right] \leq \delta_n, \quad (44)$$

where expectation is with respect to $\varphi_{1,b}$ and $\varphi_{2,b}$.

2) $\mathbf{P}^{\text{RC}}$ close to $\mathbf{P}^{\text{RB}}$: Here, we apply [12, Theorem 1] to the bin indices (F_b, J_b) , with side information $(U_{b-1}^n, W_{1,b-1}^n, X_{b-1}^n, Z_{b-1}^n)$. This allows us to conclude that if

$$\tilde{R}_F + \tilde{R}_J < H(W_1) + H(W_2|U, X, Z, W_1), \quad (45)$$

then there exists $\epsilon_n \rightarrow 0$ such that

$$\mathbb{E}_\varphi \left[\left\| \mathbf{P}_{\Sigma_{b-1}^n F_b J_b}^{\text{RB}} - \mathbf{P}_{\Sigma_{b-1}^n}^{\text{RB}} p_{F_b}^U p_{J_b}^U \right\|_{\text{TV}} \right] \leq \epsilon_n. \quad (46)$$

Since the encoder kernel in $\mathbf{P}^{\text{RC}}$, defined by (30), is equal to the corresponding kernel in $\mathbf{P}^{\text{RB}}$, and all remaining kernels coincide, [15, Lemma 17] combined with (46) gives

$$\mathbb{E}_\varphi \left[\mathbb{E}_{\mathbf{P}_{\Sigma_{b-1}^n}^{\text{RB}}} \left[\left\| \mathbf{P}_{\Lambda_b^n | \Sigma_{b-1}^n}^{\text{RC}} - \mathbf{P}_{\Lambda_b^n | \Sigma_{b-1}^n}^{\text{RB}} \right\|_{\text{TV}} \right] \right] \leq \epsilon_n. \quad (47)$$

3) *Total Variation of the Block Transitions*: Combining (44) and (47), and then marginalizing to Λ_b^n , yields

$$\mathbb{E}_\varphi \left[\mathbb{E}_{\mathbf{P}_{\Sigma_{b-1}^n}^{\text{RB}}} \left[\left\| \mathbf{P}_{\Lambda_b^n | \Sigma_{b-1}^n}^{\text{RC}} - \mathbf{P}_{\Lambda_b^n | \Sigma_{b-1}^n}^{\text{RB,perf}} \right\|_{\text{TV}} \right] \right] \leq \epsilon_n + \delta_n. \quad (48)$$

Together with (39), this shows that one block of the random-coding scheme simulates the target transition from $(U_{b-1}^n, H_{1,b-1}^n)$ to $(V_{b-1}^n, H_{2,b-1}^n, U_b^n, H_{1,b}^n)$, where

$$H_{1,b}^n = (W_{1,b}^n, X_b^n, Y_b^n, Z_b^n), \quad H_{2,b}^n = W_{2,b}^n.$$

Appending S_{b-1}^n and $\hat{S}_{b-1}^n$ through the memoryless kernels induced by $\bar{P}_S \bar{P}_{Y|XZ}$ and $Q_{\hat{S}|XZ}$ then gives (20b).

C. Removal of the Extra Common Randomness F_b

Recall that $F_b = \varphi_{1,b}(W_{1,b}^n, W_{2,b-1}^n)$. By [12, Theorem 1] applied to the bin index F_b with side information $(U_{b-1}^n, V_{b-1}^n, \Sigma_b^n)$, we have

$$\mathbb{E}_\varphi \left[\left\| \mathbf{P}_{U_{b-1}^n V_{b-1}^n \Sigma_b^n F_b}^{\text{RB,perf}} - Q_{UV}^{\otimes n} Q_{\Sigma}^{\otimes n} p_{F_b}^U \right\|_{\text{TV}} \right] \rightarrow 0 \quad (49)$$

if

$$\tilde{R}_F < H(W_2|U, V). \quad (50)$$

$W_{1,b}^n$ is contained in Σ_b^n , so the remaining conditional entropy of the binned pair is $H(W_2|U, V)$. Similarly, applying the same theorem with side information $(S_{b-1}^n, \hat{S}_{b-1}^n, \Sigma_b^n)$ proves that

$$\mathbb{E}_\varphi \left[\left\| \mathbf{P}_{S_{b-1}^n \hat{S}_{b-1}^n \Sigma_b^n F_b}^{\text{RB,perf}} - Q_{S\hat{S}}^{\otimes n} Q_{\Sigma}^{\otimes n} p_{F_b}^U \right\|_{\text{TV}} \right] \rightarrow 0 \quad (51)$$

whenever

$$\tilde{R}_F < H(W_2|S, \hat{S}). \quad (52)$$

Since, on expectation, the pmfs $\mathbf{P}^{\text{RC}}$ and $\mathbf{P}^{\text{RB,perf}}$ have vanishing TV distance on the relevant marginals, the limits in (49) and (51) also hold for $\mathbf{P}^{\text{RC}}$. By applying [12, Lemma 4] to the sum, over all active blocks, of the TV distances in (49), (51), and the corresponding Slepian–Wolf reconstruction-error probabilities, there exists a realization f^B of F^B such that all these summed quantities vanish:

$$\sum_{b=2}^{B+1} \left\| \mathbf{P}_{U_{b-1}^n V_{b-1}^n \Sigma_b^n | F^B = f^B}^{\text{RC}} - Q_{UV}^{\otimes n} Q_{\Sigma}^{\otimes n} \right\|_{\text{TV}} \rightarrow 0, \quad (53)$$

$$\sum_{b=2}^{B+1} \left\| \mathbf{P}_{S_{b-1}^n \hat{S}_{b-1}^n \Sigma_b^n | F^B = f^B}^{\text{RC}} - Q_{S\hat{S}}^{\otimes n} Q_{\Sigma}^{\otimes n} \right\|_{\text{TV}} \rightarrow 0. \quad (54)$$

Thus, F^B can be fixed. After fixing $F^B = f^B$, block b uses the carry Σ_{b-1}^n to produce V_{b-1}^n for the UV -coordination marginal and $\hat{S}_{b-1}^n$ for the $S\hat{S}$ -sensing marginal, and to refresh the carry to Σ_b^n . The bounds (53) and (54) show that this refreshed carry is asymptotically distributed as $Q_{\Sigma}^{\otimes n}$ and is independent of the corresponding UV - or $S\hat{S}$ -block. Hence the same block-propagation and triangle-inequality argument as in (20)–(21) applies. The termination block $b = B + 1$ is handled in the same way after marginalizing the final carry. This gives

$$\|P_{UVN} - Q_{UV}^{\otimes N}\|_{\text{TV}} \rightarrow 0, \quad \|P_{S\hat{S}N} - Q_{S\hat{S}}^{\otimes N}\|_{\text{TV}} \rightarrow 0.$$

D. Proof of the Rate Condition (7)

Combining the rate constraints (31), (32), (50), and (52), together with the fact that after fixing F^B only J^B consumes external common randomness, i.e., $R_C > \tilde{R}_J$, Fourier–Motzkin elimination of $(\tilde{R}_F, \tilde{R}_J)$ gives (6) and (7). The equality case follows by using the remaining channel uses to convey an arbitrarily small-rate refinement index, which makes the Slepian–Wolf inequality strict without affecting (32).

APPENDIX B

DETAILS FOR THE ESTIMATOR CONVERSE

For each blocklength N and time $i \in [1 : N]$,

$$(U^N, C, X_{\sim i}^{N'}, Z_{\sim i}^{N'}, \hat{S}_i) - (X_i, Z_i) - S_i. \quad (55)$$

Given (X_i, Z_i) , the state S_i is conditionally independent of the source sequence, the common randomness, and the remaining channel inputs and feedback symbols; moreover, $\hat{S}_i$ is generated from these variables. Hence

$$P_{S_i \hat{S}_i X_i Z_i}^{(N)} = P_{\hat{S}_i | X_i Z_i}^{(N)} P_{S_i X_i Z_i}^{(N)} = P_{\hat{S}_i | X_i Z_i}^{(N)} P_{X_i}^{(N)} \bar{P}_S \bar{P}_{Z|XS}. \quad (56)$$

Averaging over i , and multiplying and dividing by $\frac{1}{N} \sum_{i=1}^N P_{X_i Z_i}^{(N)}(x, z)$, define

$$P_{\hat{S}|XZ}^{(N)}(\hat{s}|x, z) \triangleq \frac{\frac{1}{N} \sum_{i=1}^N P_{\hat{S}_i | X_i Z_i}^{(N)}(\hat{s}|x, z) P_{X_i Z_i}^{(N)}(x, z)}{\frac{1}{N} \sum_{i=1}^N P_{X_i Z_i}^{(N)}(x, z)}, \quad (57)$$

with arbitrary definition when the denominator is zero. Then

$$P_{S_T \hat{S}_T X_T Z_T}^{(N)} = P_X^{(N)} \bar{P}_S \bar{P}_{Z|XS} P_{\hat{S}|XZ}^{(N)}, \quad (58)$$

where

$$P_X^{(N)}(x) = \frac{1}{N} \sum_{i=1}^N P_{X_i}^{(N)}(x). \quad (59)$$

Taking the limit along the subsequence $\{N_i\}$ in the converse of Theorem 1 gives

$$\tilde{Q}_{S\hat{S}XZ} = Q_X \bar{P}_S \bar{P}_{Z|XS} Q_{\hat{S}|XZ} = \tilde{Q}_{SXZ} Q_{\hat{S}|XZ}. \quad (60)$$

Therefore, on the support of $\tilde{Q}_{SXZ}$,

$$Q_{\hat{S}|XZS} = Q_{\hat{S}|XZ}, \quad (61)$$

with arbitrary definition on zero-probability points.